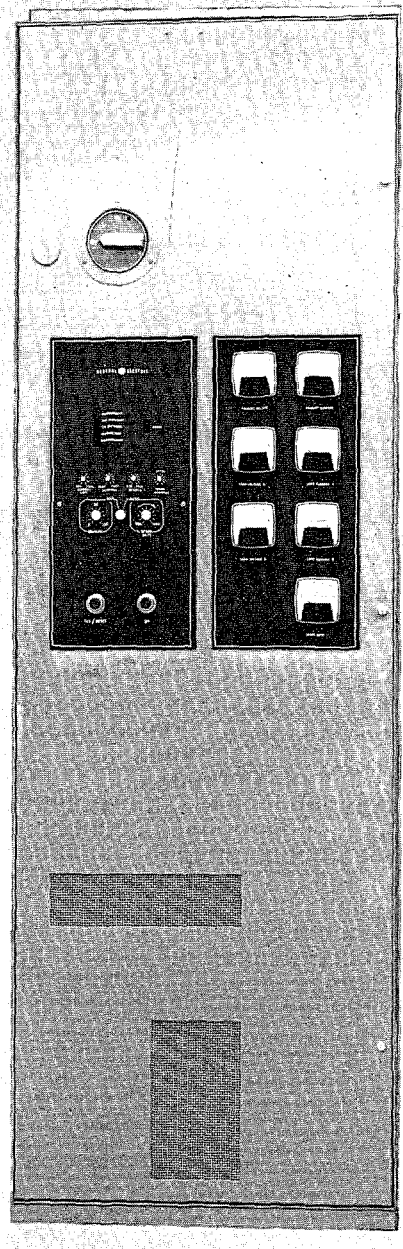
INSTRUCTIONS

GEK-40743

IC100
ELECTROSTATIC PRECIPITATOR CONTROLLER



GENERAL ELECTRIC

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These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connectron with installation, operation or maintenance. Should further information be dewed or should particular problems arise which ore not covered sufficiently for the purchaser's purposes, the matter should be referred to the General Electric Company.

IC100
ELECTROSTATIC
PRECIPITATOR CONTROLLER

1.0 GENERAL

The primary purpose of an electrostatic precipitator control is to provide maximum excitation within the precipitator consistent with acceptable spark levels. In addition, the control must recognize a spark, extinguish it, and reapply precipitator excitation,

The General Electric IC100 Electrostatic Precipitator Control (Fig. 1.1) has been designed to operate with a high-voltage transformer-rectifier so as to accomplish maximum precipitator efficiency. This is accomplished by continually seeking the sparking or arc-over voltage level of the precipitator unit and resetting the precipitator excitation just below this level. Without sparking, the control will continually increase precipitator ***excitation*** until rated limits are reached. When a spark or arc occurs within the precipitator, the control's SCR's (Silicon Control Rectifier) block primary current flow until the arc ***is*** cleared and then rapidly reapply power to a level just below the point where the last spark occurred.

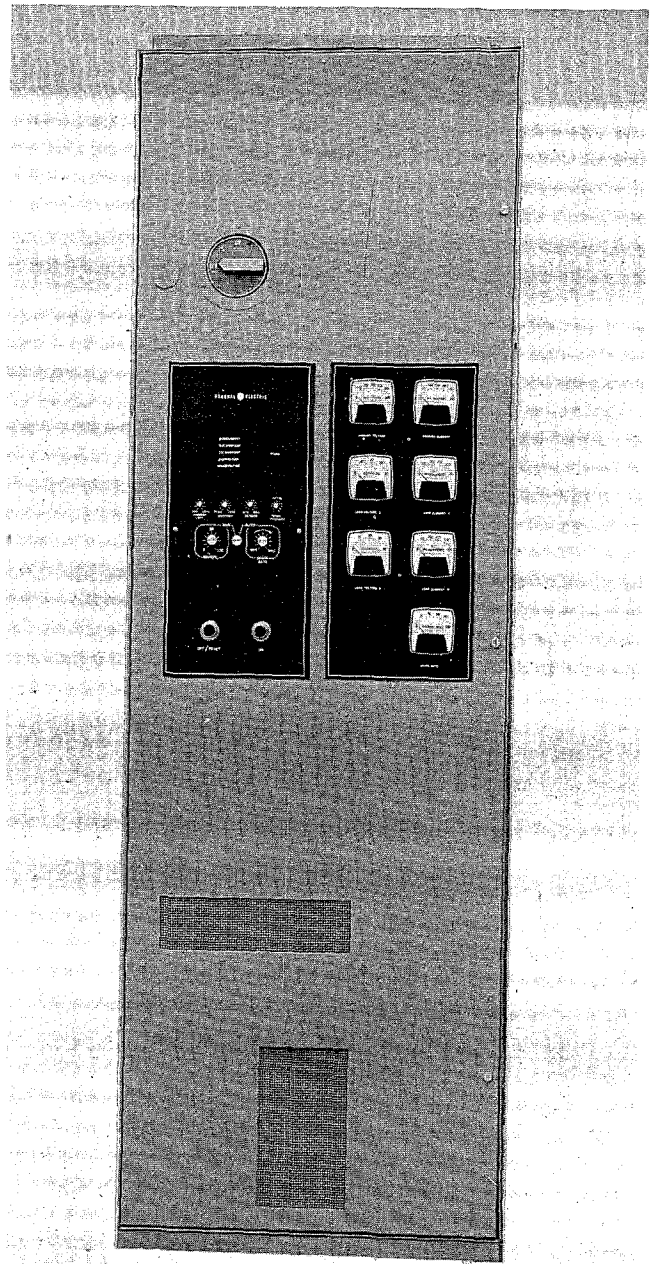


Fig. 1.1 IC100 controller

2.0 IDENTIFICATION/DESCRIPTION

An I. D. nameplate ***is*** attached to the device panel (Fig. 2.6) of each controller. The nameplate includes the controller catalog number, the specific diagram number and the controller electrical ratings.

The controller utilizes modular construction consisting of a meter panel, controlpanel, a circuit breaker panel, a device panel and an SCR and reactor assembly. These modules are wired using wire harnesses with 'plug on" edge connectors and terminal board terminations. The controller shell is a bolted assembly with bolt-in modules to minimize component removal and replacement effort. The following paragraphs describe each of the panel module :

2.1 METER PANEL

The meter panel (Fig. 2.1) is a removable bolt-on panel containing individual removable meters used for monitoring *primary and load transformer-rectifier voltage and current and precipitator *spark rate. The meters are standard 1.0 milliamp dc movements with scales selected for the controller requirements. The electronic cards condition metering signals for the 1.0 milliamp movements. The *primary and load current meters are scaled based on the controller rating. The *primary voltage meter is always scaled O-600 volts ac, the load voltage meter(s) are always scaled O-80 kilovolts and the *spark rate meter is always scaled O-100 sparks per minute regardless of rating or line voltage. Meters and driver electronics are protected if an open circuit or short from meter terminal to terminal or terminal to ground occurs. A maximum of 15 volts dc is the highest voltage on the meter panel. The meter panel is electrically connected to the PRGA electronic card through edge connector 6C as labelled on the card. The connecting wire harness has wire number markers at each end (at the meters, terminals and the card connectors) and detailed on elementary drawing 205B7590. (For remote metering see Paragraph 2.4).

*Optional

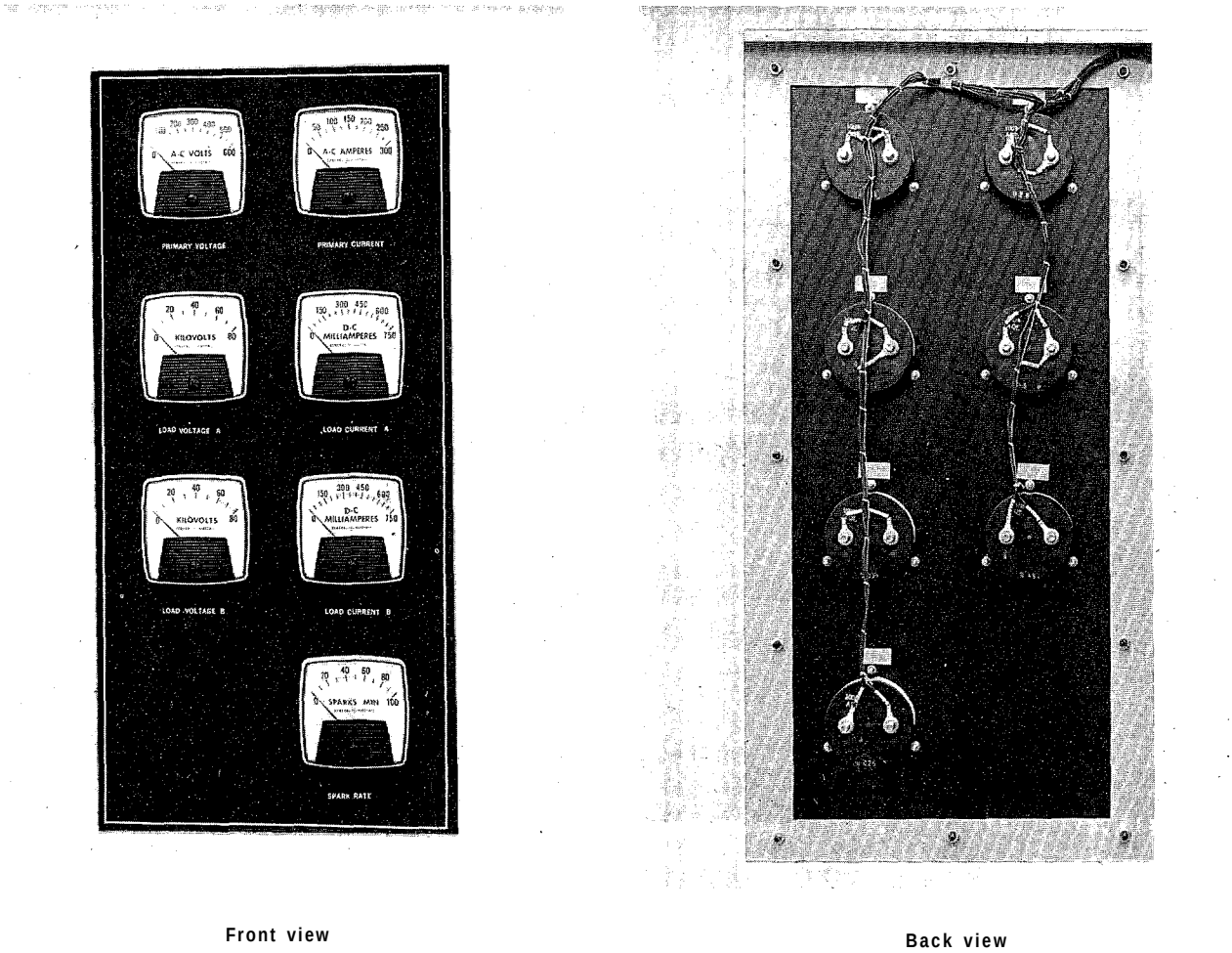
2.2 CONTROL PANEL

The control panel (Fig. 2.2) is a removable bolt-on panel containing:

1. The electronic card (PREA).

2. The program, adjustment, and alarm indication/trip card (PRGA).

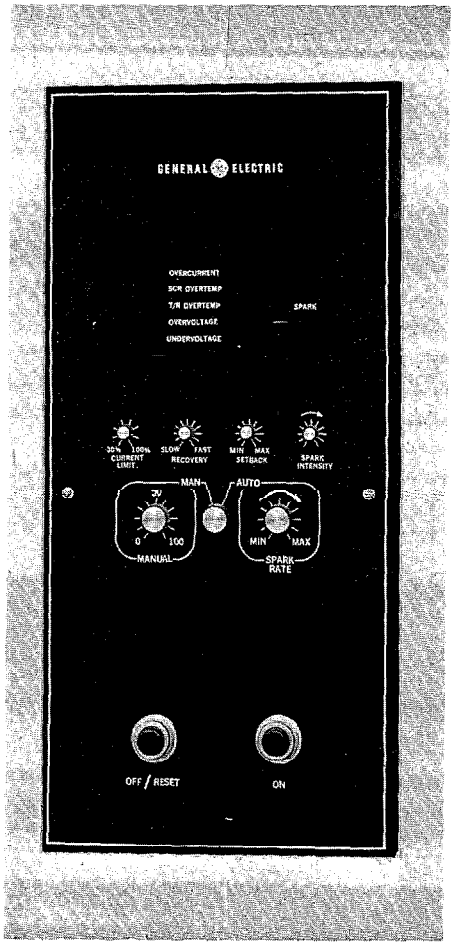
3. Illuminated on and off/reset push buttons.



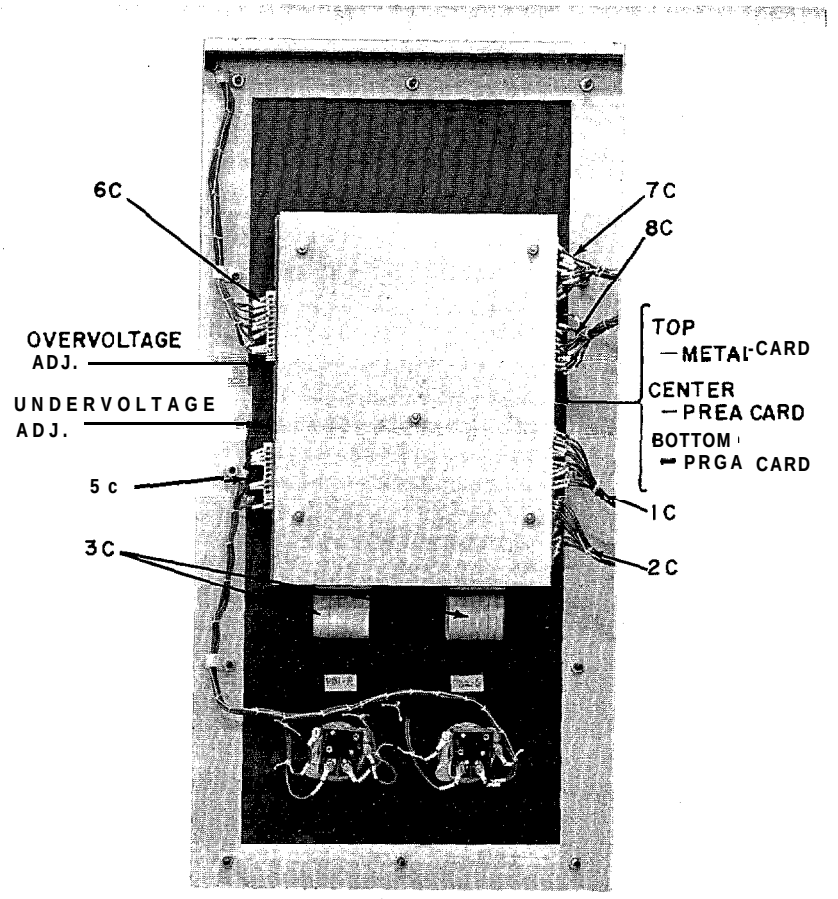
Front view

Back view

Fig. 2.1 Meter panel



Front view



Back view

Fig. 2.2 Control panel

4. Spark light.

5. A metal cover plate for card shielding.

6. A Plexiglass bolted cover (not shown), mounted over the adjustment knobs.

The PRGA card contains male edge connectors for all input and output electrical connections and 34-pm male connectors for electrical interface to the PREA card. The PREA has two 34-pin male connectors which provide allelectrical connections via the ribbon cables. The right connector (Fig. 2.2) has two rows of pins, Row A and B, numbered 1 thru 17. The left connector has pin rows A and B and numbered 18 thru 34.

The PRGA card is constructed in such away that light-emitting diodes (LED) lights and control adjustments are mounted directly on the card. Control adjustments extend through the door for external front adjustment and the LED lights are positioned to be viewed through a red transparent window. The following control adjustments and selections are provided to control applied power to the high voltage transformer-rectifier: manual/auto selection, spark rate, current limit, recovery setback and spark intensity.

The following control trip and indications are provided: Overcurrent, SCR overtemperature, *T/R overtemperature, *overvoltage, *undervoltage, and on/off illuminated push buttons. For control adjustment, see Paragraph 8.3.

2.3 CIRCUIT BREAKER PANEL

The circuit breaker panel (Fig. 2.3) consists of a two-pole thermal-magnetic circuit breaker, line shield and breaker support. The CB is selected based on controller primary amperes and short circuit rating required. The CB's thermal trip provides over-current protection and is selected for

*Optional

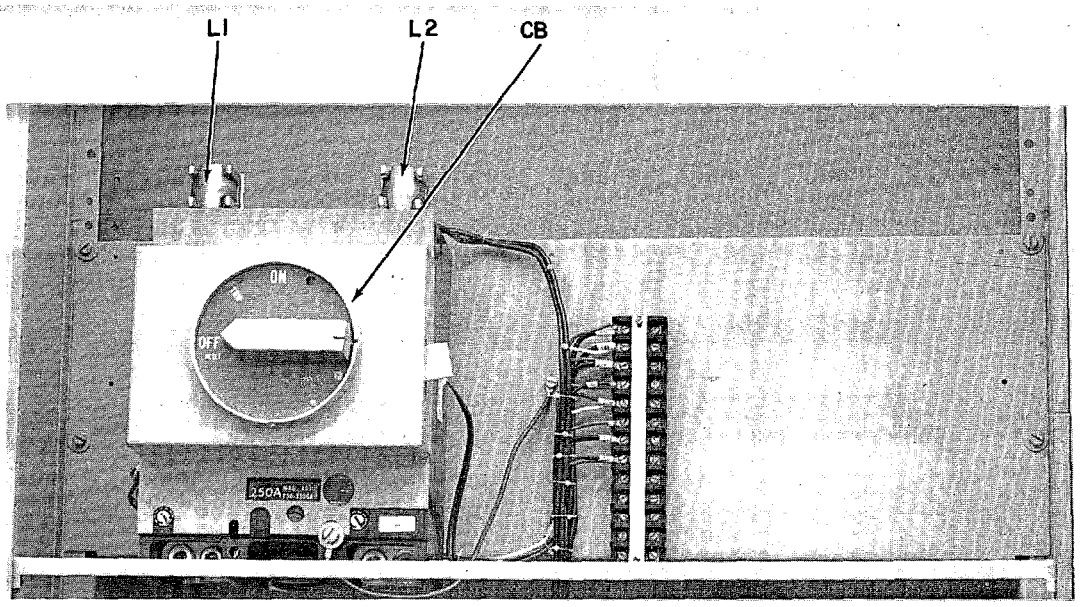


Fig. 2.3 Circuit breaker panel

125 percent of control primary rated current. The magnetic trip is an instantaneous trip set at 6X full load current and provides short circuit protection.

Incoming line power connects to the line side of the circuit breaker (Fig. 2.3). Power connections to the transformer-rectifier connects to terminals TR1 and TR2, (Fig. 2.6 and 2. 7).

Provisions are supplied for a Kirk lock or Superior lock mechanism (Fig. 2.4) to be mounted on the CB handle assembly for locking the circuit breaker in the OFF position. A removable round cover plate covers the door 'cutout for the lock barrel to extend through. The thick gasket used with the coverplate must be slipped over the lock barrel to provide a door seal. The coverplate may be discarded. Detail keylock information is shown on drawings 278A5011P1, 278A5011P2, and 272A5630P1.

The circuit breaker handle mechanism (Fig. 2.5) mechanically interlocks with the controller door preventing unauthorized opening of the door when the handle is on. The CB may be padlocked in the ON or OFF position by extending the locking hasp and installing up to three padlocks. With padlocks removed, access to the control can be gained with power on by turning the defeater "A" counter-clockwise.

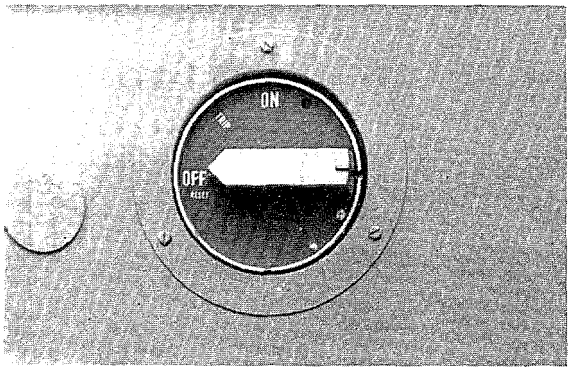
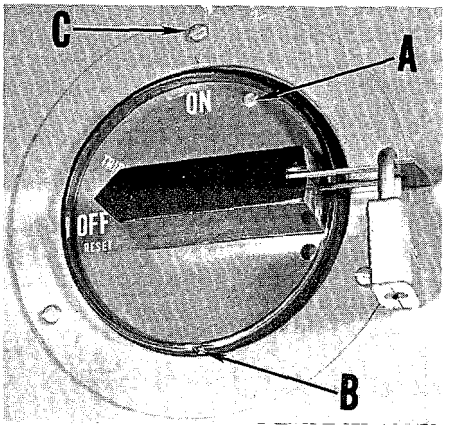


Fig. 2.4 **Provision for key lock mechanism**



A. Defeater
B. **lock-on provision**
C. Adjustable floating ring screws

Fig. 2.5 Operating handle with slide extended for lock OFF provision

2.4 DEVICE PANEL

The device panel (Fig. 2.6) consists of the incoming control terminal boards, input-output conditioning circuitry, a 15 volts dc power supply, *contactor, fan and fuse protection.

All customer control wire interconnections terminate on the device panel. Terminal board connections are provided on the right-hand side of the PIFA card for the transformer-rectifier (T/R) control interconnections and on the right-hand side of the *PRIA card for all remote control and indications. Card terminal boards are rugged 600 volts rated and contain screw followers.

Input signals from the T/R are filtered and conditioned in the PIFA card and then routed via a cable harness to the door mounted electronics cards. These signals include:

1. T/R secondary voltage
2. T/R secondary current
3. *T/R overtemp indication
4. Remote trip.

*Optional

Output signals for T/R status are routed through the PIFA card. Output signals are:

1. T/R power on
2. T/R power off

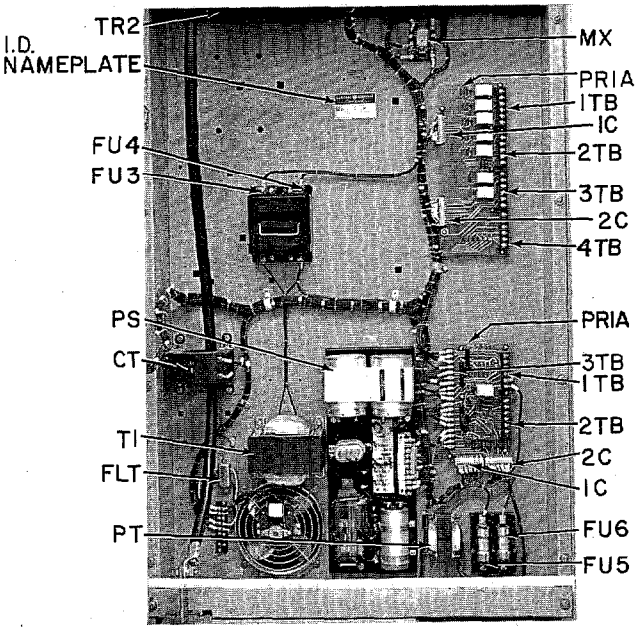
Input signals for remote control to the *PRIA card are:

1. Manual adjust
2. Manual/auto selection
3. Local/remote selection

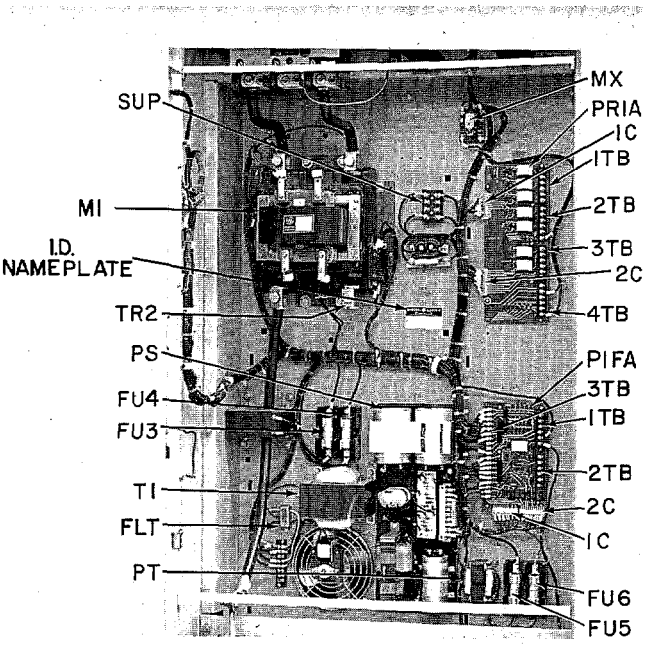
Output metering signals for remote indicationon the *PRIA card *are:*

1. Primary voltage
2. Primary current
3. Secondary voltage
4. Secondary current
5. Spark rate

Metering signals are for 1.0 milliamp dc movement meter, generated from O-10 volts dc signals



Standard



With optional contactor

Fig. 2.6 Device panel

through a precision 10K-ohm resistor mounted on the *PRIA Card. Remote meter scales must be the same as the controller meter scales.

Meter signals are protected if short-circuited or open-circuited. Outputs for remote use on the *PRIA card for the following alarm conditions are provided in the form of relay contacts:

1. Over current
2. SCR overtemperature
3. T/R overtemperature
4. Undervoltage
5. Overvoltage

Outputs for remote on and off/reset pushbuttons and relay contacts for on and off lights are provided. All relay contacts for remote use are rated 2.0 ampere at 28 volts dc or 120 volts ac.

2.5 SCR REACTOR ASSEMBLY

The SCR Reactor assembly (Fig. 2. '7) is a sealed, bolted, enclosure containing the current limiting reactor, SCR heat sink assy., two ventilation fans and the SCR gate driver card. These devices are bolt-in and front removable. Power terminal (TR1) on the right hand side provides one of two outgoing power connections (Fig. 2.7).

Forced air circulation for this enclosure enters the front top grid cutout, flows through the SCR heat sink fans, over the reactor and out the bottom grid cutout. In addition to this air flow, on higher rated controllers, a heat exchanger is provided to allow ambient air to enter in the top of the controller, passing through heat exchanger tubes to the top of the SCR-reactor enclosure, through the forcing fans and out the lower front grid cutout.

The SCR gate driver card (PGDA) is requiredon controllers with 380 volts ac or 460 volts ac line voltage. The PGDB card is required on 5'75 volts ac controllers.

The SCR gate driver card is mounted behind a removable metal protective shield. The shield provides protection against accidental contact with the line supply voltage as well as providing electrical shielding.

Input signals to the gate driver card are connected through a cable harness to the edge connector 1C located on the right of the gate driver card. Gutput signals to drive the SCR gates and 115 volts ac power for the reactor fans connect through the 1TB terminal board on the top of the card.

*Optional

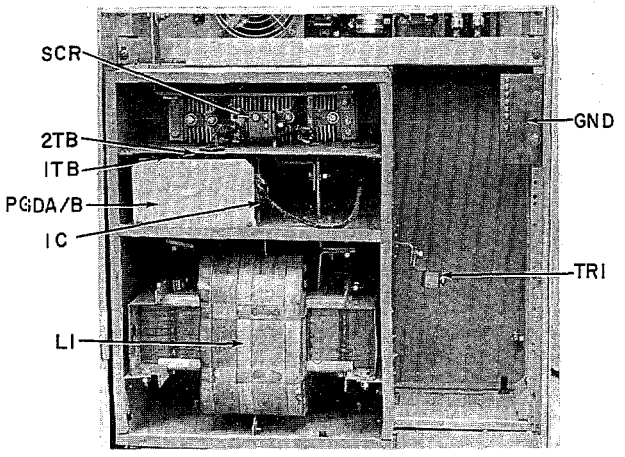


Fig. 2.7 SCR reactor assembly

3.0 SPECIFICATIONS

3.1 DIMENSIONS (Reference Fig. 3.1.)

3.2 AMBIENT TEMPERATURE

During Operation	Storage
0 to 40 C Standard	**-25 c to 70 c**
0 to 50 C Optional	**-25 C** to 70 C
95-percent Maximum Humidity	95-percent Maximum Humidity
No Condensation	No Condensation

3.3 POWER SOURCE

380 Volts ac, ±5 percent, 1 Phase, 50 Hz
460 Volts ac, ±5 percent, 1 Phase, 60 Hz (Standard)
575 Volts ac, ±5 percent, 1 Phase, 60 Hz

3.4 INPUT SIGNALS

A. TR Secondary Current Feedback

B. TR High Voltage Signal Feedback

C. *TR Overtemperature - Normally closed contact from T/R.

Rating Requirement:

0. 1 Amp @ 90 Volts dc

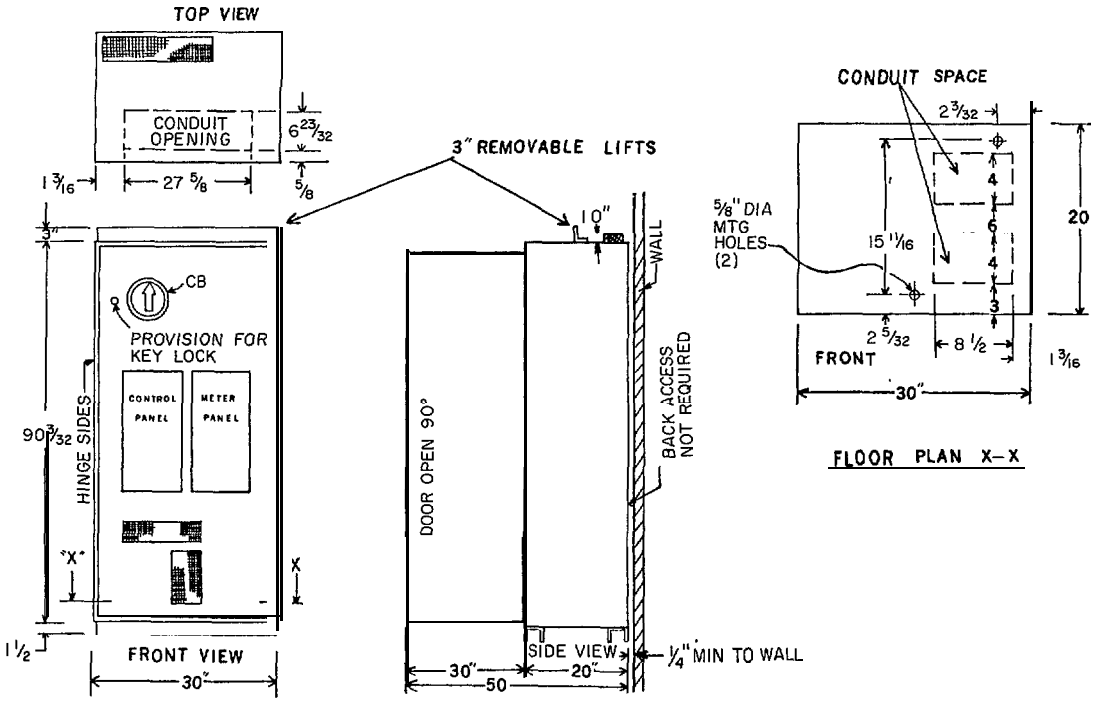


Fig. 3.1 Outline dimensions of precipitator controller

D. *Remote Control Rating Requirement : (Reference Fig. 5.2)

1. Remote ON/OFF Push button - 0.1 Amp @ 30 Volts dc
2. Remote trip- 0.1 Amp @ 30 Volts dc
3. Manual Adjust - 10K ohm Linear Potentiometer 1/2 watt
4. Manual/Auto Select - 0.1 Amp @ 30 Volts dc
5. Local/Remote Select- 0.1 Amp @ 30 Volts dc

3.5 OUTPUT SIGNALS

A. *Alarm Contacts-Rating;
2 Amperes *at* 120 Volts ac resistive load
2 Amperes at 28 Volts dc resistive load

SCROT o c TROT u v o v	Close on Alarm
ON OFF	Close for power on/off (For Remote On/Off lights)

B. *CB---(On and Off Status)
C. *M ---(Energized and De-energized Status)

ie: NO and NC AUX. Interlocks are available on CB or M, but not both.

*Optional

D. Metering - 1 Milliamp dc movement (Reference Interconnection diagram, Fig. 5.2)

3.6 CONTROL ADJUSTMENT RANGE

(Reference Fig. 3.2)

A. Current Limit--0-100% rated current

B. Recovery --14 - 70 ms, "OFF-TIME"
10 - 100 ms. "FAST-RAMP"

C. Step Back --0.5 - 10% of rated current

D. Spark Rate --0.5 - 10 minutes from zero to rated current

E . Intensity --Minimum to maximum

F. Manual --0 - 100% rated current

G. *Undervoltage--Trimpot - 20-175 Volts ac for 1/2, 1 or 2 min. time delay

H. *Overvoltage --Trimpot - 330-650 Volts ac

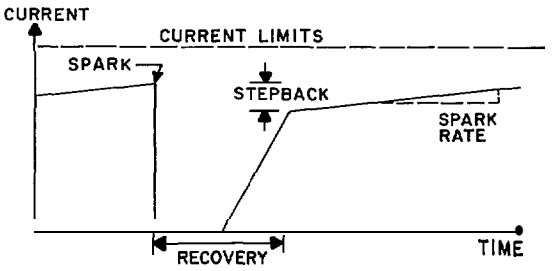


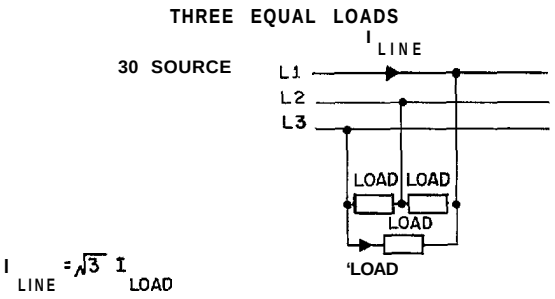
Fig. 3.2 Adjustment *pictorial*

3.7 FORM FACTOR

The transformer and control ratings are based upon a 1.19 form factor for secondary dc voltage and a 1.2 form factor for secondary dc current as applied to a modified resistive load. These form factors are the ratio of root mean square values to average values. The kVA rating is therefore the product of the two form factors, or 1.43 times average secondary dc kilovolts and average dc milliamperes.

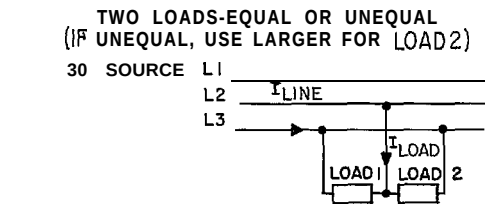
3.8 LINE CURRENT CALCULATIONS

(Reference Fig. 3.3)



$$I_{LINE} = \sqrt{3}\, I_{LOAD}$$

$$I_{LOAD} = \frac{KV \times MA \times 1.43}{T/R\ PRI.\ VOLTAGE}$$



$$I_{LINE} = \left[\left(I_{LOAD\ 1} + 0.5 I_{LOAD\ 2} \right)^2 + \left(0.866\ I_{LOAD\ 2} \right)^2 \right]^{1/2}$$

$$I_{LOAD} = \frac{KV \times MA \times 1.43}{T/R\ PRI.\ VOLTAGE}$$

Fig 3.3 Line current loads

4.0 RECEIVING, HANDLING, STORAGE

4.1 RECEIVING

Before leaving the factory, the Precipitator controller is given a final mechanical and electrical inspection and is packed in keeping with best practices for electrical and electronic equipment.

Upon receipt of any apparatus, an immediate inspection should be made for any damage or loss of equipment in transit. Should damage or missing material be noted, a claim should be immediately filed with the carrier and the nearest office of the General Electric Company. Information such as description of damage, shipping crate numbers, requisition numbers, etc., should accompany the claim.

4.2 HANDLING

Controllers should always be shipped in an upright position during all handling.

Never attempt to jack, lift, or move the equipment at points other than the lifting angle, lifting lugs or floor sills. Use two or more lifting ropes or cables to distribute weight more evenly. Pinch-bars, pipe rollers, or slings are useful implements for handling equipment; but care must be taken to maintain distributed loading and to always apply leverage at the floor sills and/or lifting angle.

4.3 STORAGE

NOTE: *If it is necessary to store the equipment for any length of time, the following precautions should be taken:*

1. Uncrate equipment.

2. Store in a clean, dry area at moderate temperature. Cover with a suitable canvas or heavy-duty plastic cover to prevent entrance of foreign material.

3. If equipment must be stored in cool or damp areas, not only should the equipment be completely covered, but heat should be provided to prevent condensation of moisture in the equipment. Energize space heaters (if furnished in the equipment) or place a standard 120-volt lamp rated 150 watts inside the bottom of each controller.

Before any installation work is done, consult all drawings furnished by General Electric Company as well as all applicable contract drawings for the particular installation. Particular attention should be given to the controller outline drawing in relation to existing or planned conduits, busways, etc..

Care should be taken to plan for any future conduit entrance in advance of control center installation.

5.0 INSTALLATION

5.1 MOUNTING

(Reference Fig. 3.1)

5.2 CONDUIT OPENINGS

Conduit openings are provided for the top or bottom cable entry. (Reference Fig. 3.1.) All panels contain removable conduit plates in the bottom for control and power cable entry.

5.3 INCOMING POWER CONNECTIONS

Panels are fabricated from sheet steel for single or *multiple-panel line-ups. *Multiple -panel line ups may contain 3-phase bus bar.

Incoming power connects directly to the circuit breaker for single panels or to the 3-phase horizontal bus bars for multiple-panel line up. Solderless compressionlugs are supplied for all incoming and outgoing power cabling. Information on bus rating, short-circuit bracing and 1 -phase loading for bused line-ups are provided for each job on the line-up outline drawing. Balanced loading for single-phase connections are made to the 3 -phase bus from the front, left to right: Phase A-B, A-C, and B-C unless otherwise specified.

5.4 VENTILATION REQUIREMENT

Forced ventilation enters the front cutout and on some ratings the top cutout to allow the control to be mounted flush against the wall. Ventilation cutouts must not be restricted.

5.5 CONTROLLER GROUNDING

Controller grounding is essential for safety and proper control operation. Ground cabling should connect the controller to the transformer-rectifier and to the precipitator.

A copper ground pad (approximately 8.2 5 inches x 3 inches x 0.25 inches) is provided for proper controller grounding. The pad (Fig. 5.1) has a 0.44 inch diameter hole for bolting the ground cable.

5.6 INTERCONNECTING WIRING

Reference interconnection diagram. (Reference Fig. 5.2).

*Optional

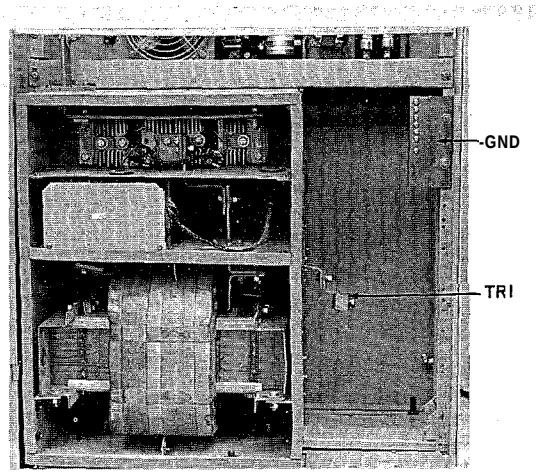


Fig. 5.1 Ground pad

NOTE: ***This interconnection diagram*** *includes optional control functions.* ***Each job will include an interconnection diagram showing its particular*** *requirements .*

1. All interconnecting control wiring should be No. 16 AWG, or larger, twisted and shielded. The shield should be grounded at the controller end only. Controlwiring should be run at least 18 inches from power cables.

2. Use power cable size corresponding at least to nameplate ratings of controller.

3. Use thermally suitable cable where subjected to above normal ambient.

4. Use wire numbers, or other identificationfor all interconnecting wire and cable.

5. Include extra control wires between devices and panels when additional control functions might later be considered.

6. Protect all interconnecting wire and cable from abrading on sharp or rough edges.

'7. Holes are provided for cable clamps to support incoming wires on the CB panel and vertically along the right side of the controller.

6.0 POWER APPLIED AND STARTUP

1. Visually inspect all equipment prior to applying power. Verify and tighten incoming and outgoing power connections. Verify that power source voltage agrees with the control I. D. nameplate voltage, (Reference Fig. 2.6.)

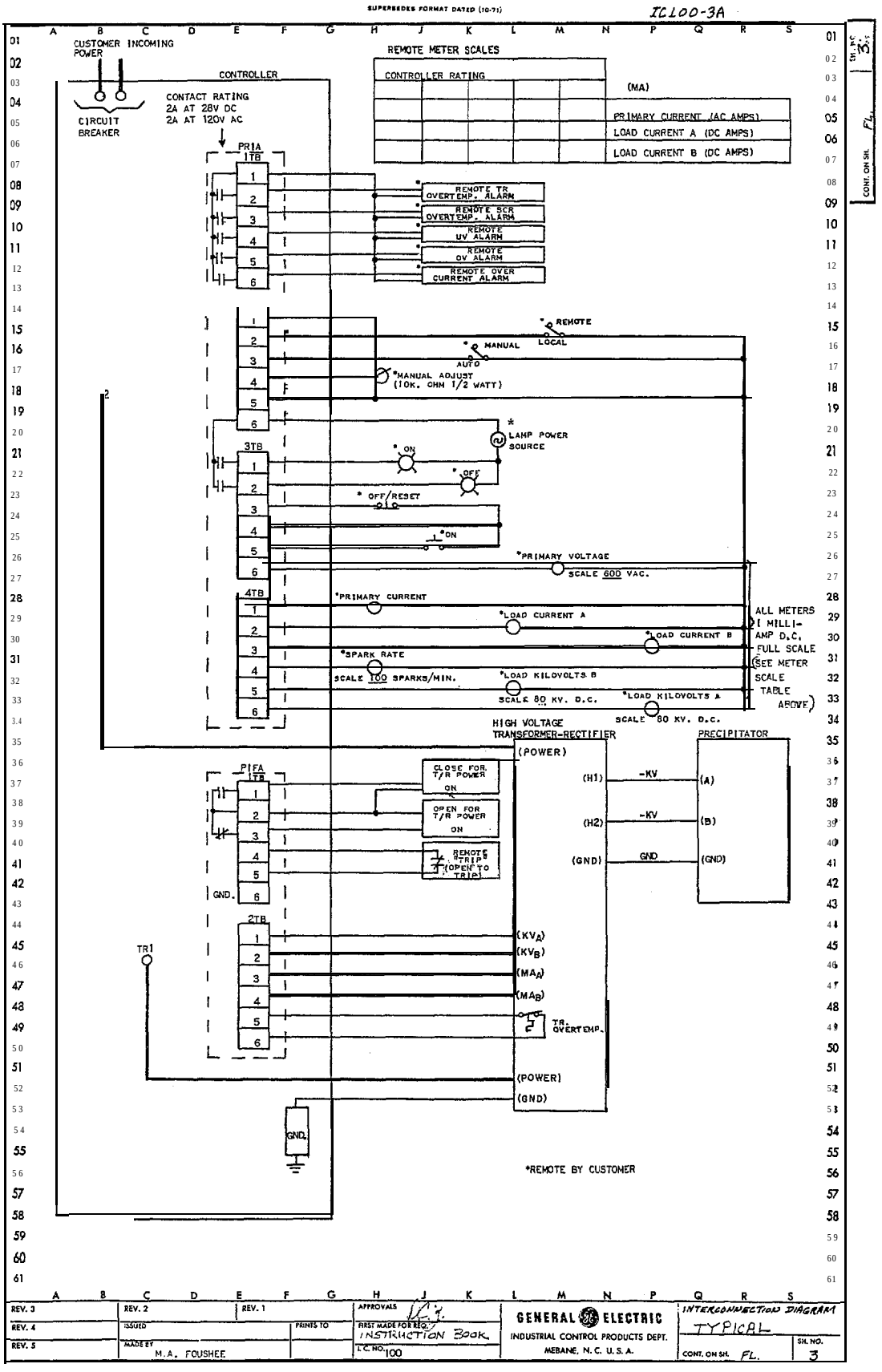


Fig. 5.2 *Interconnection* **diagram**

2. Before power is applied, measure resistance with an ohmmeter from each of the following terminals to the ground connection in the control to verify T/R secondary current feed-back ground return.

PIFA 2TB-3 to ground lug to be between 0.5 and 2 ohms

PIFA 2TB-4 to ground lug to be between 0.5 and 2 ohms

3. Set controls located on panel front as follows :

(Reference Fig. 2.1)

Manual/Auto Switch - Manual
Spark Rate Potentiometer - Midpoint
Manual Potentiometer - Fully CCW
Current Limit Potentiometer - 100 Percent (Percent of Panel Rating)
Recovery - One Third CW
Setback - Midpoint
Spark Intensity - Fully CCW (Maximum Sensitivity to sparks)

4. Close Main Circuit Breaker.

5. Note that the green "OFF" light is lit. Clear any alarm indication by depressing the "OFF/RESET" PB. Note that no alarm lights are lit. Depress the "ON" PB. Note that the green light goes out and the red light comes on.

5. Increase precipitator current by increasing "Manual" potentiometer clockwise. Observe metering and note an increase in current and voltage. Return Manual adjustment CCW. If anundervoltage alarm is included, the alarm will occur in approximately one minute. (Reference undervoltage under Alarm and Trip Indications, Paragraph 6.1).

7. Select automatic control by placing the Auto/Manual switch to "Auto" and note primary current gradually increases to the current limit setting or sparking. When a spark isdetected, the sparklight will flash and the control will automatically "setback" by an amount set by the setback adjustment.

8. Stop and restart the control noting no alarms and a return to normal operation.

6.1 ALARM AND TRIP INDICATIONS

(Reference Fig. 2.2)

The following is a complete list of alarms and trips. The particular job elementary diagram specifies which alarms, alarms and trips, or combinations are included with each job.

Overcurrent - This alarm indication and trip is included on all controls. "Overcurrent" lights and the control disconnects power from the transformer rectifier if maximum rated primary current is exceeded. Maximum rated primary current is tabulated on the elementary and controller nameplate. (Reference Fig. 2.6). By depressing the OFF/RESET push button, the alarm light and trip are reset.

SCR Overtemperature - This alarm indication and trip is included on all controls. The alarm occurs if a normally closed contact in the "SCR heat sink" or the "reactor core" opens for an overtemperature condition or relay K1 on the PGDA/PGDB card energizes if the primary current is not symmetrical. The red alarm indication labeled "SCR OVERTEMPERATURE" lights and the control disconnects power from the transformer rectifier. By depressing the OFF/RESET push button, the alarm light and trip are reset.

*T/R Overtemperature - A normally closed contact in the transformer rectifier (T/R) opens if an overtemperature condition occurs. The red alarm indication labeled "T/R OVERTEMPERATURE" lights and if a trip is furnished the control disconnects power from the transformer rectifier. By depressing the OFF/RESET push button, the alarm light and trip are reset.

*Overvoltage - The transformer rectifier primary voltage is monitored through transformer PT. If an overvoltage condition occurs. the red alarm indication labeled "Overvoltage" will light and if a trip is furnished, the control disconnects power from the transformer rectifier. By depressing the OFF/RESET push button, the alarm light and trip are reset. Overvoltage is factory set at 100 percent of rated line voltage. This setting is field adjustable from 330 volts ac (CCW) to 650 volts ac (CW). With the control door open and facing the inside of the door, the adjustment potentiometer is located on the left side of the PRGA card. The card is door mounted, has two potentiometers, with the top one the overvoltage adjustment. (See Fig. 2.2).

*Undervoltage - The transformer rectifier primary voltage is monitored through transformer PT. If an undervoltage condition exists for one minute, the red alarm indication labeled "UNDERVOLTAGE" will light and if a trip is furnished, the control disconnects power from the transformer rectifier.

*Optional

(Options of 1/2 or 2 minutes are available.) By depressing the OFF/RESET push button, the alarm light and trip are reset. The undervoltage is factory set at 10 percent of rated line voltage. This setting is field adjustable from 20 (CCW) to 1'75 volts ac (CW). With the control door open, the adjustment potentiometer is located on the left side of the PRGA card. The card is door mounted, has two potentiometers, with the bottom one the undervoltage adjustment. (Reference Fig. 2.2).

6.2 SPARK INDICATORS

Spark - The spark indicator is included on all panels. It flashes whenever the control senses a spark in the precipitator.

6.3 CONTROL ADJUSTMENTS

The following adjustments are always included and are accessible from the front of the control panel. A functional description of each control adjustment and a typical setting is given. The precipitator manufacturer may align these settings based on the precipitation process.

Manual/Auto Switch - The primary difference between manual operation and automatic operation is that in manual, the precipitator excitation is limited, by the manual adjustment. The spark intensity, recovery, stepback, and the slow ramp features' are identical in the manual and auto mode. In the manual mode, secondary current is limited to the manual setting. The slow ramp, set by the spark rate adjustment, will ramp to the manual settingor sparking. Manual is primarily intended for initial start-up or maintenance and auto mode is used in normal operation.

Manual - With the Auto/Manual switch in manual mode, the control excitation can be manually adjusted from zero (CCW position) to the current limit setting. If the manual setting is left at zero and an undervoltage trip option is included, the control will trip in approximately one minute. (Reference "Undervoltage" under Alarm and Trip Indications. (Paragraph 6.1.)

Spark Rate - The Spark Rate adjustment sets the rate of sparking by setting the rate of increased excitation between sparks. The adjustment range is from one-half to ten minutes to ramp from zero to rated current. A more CW setting will cause a faster spark rate.

*Optional

Current Limit - The Current Limit is adjustable from 0 percent to 100 percent of primary rated current. Primary rated current is tabulated on the particular job elementary diagram and on the controller nameplate. (Reference Fig. 2.6). The Current Limit setting limits current in auto or manual mode. Sparkingwill often occur before current limit is reached and 100 percent of the rating may only be obtained on a dry or no load precipitator condition. Less than 100 percent of the current limit range may be desirable when recommended by the precipitator manufacturer.

Recovery - The Recovery Adjustment governs the SCR "Offtime" and "Turn On" rate following a spark. The recovery range of adjustment is labeled slow (CCW) to fast (CW). The range of "Offtime" is adjustable from 14 (fast) to 70 (slow) milliseconds and the "Turn On" rate from 10 (fast) to 100 (slow) milliseconds from zero to rated current. Normally a one third clockwise setting is a good starting point. This adjustment should be based on the precipitator process and characteristics. If restriking or heavy sparking occurs a slower (CCW) setting may be required. If restriking or sparking is not a problem, a faster (CW) setting may be desirable.

Setback - The Setback is defined as the difference between the excitation when a spark occurs compared to the excitation after that spark and recovery. The excitation level at the time a spark occurs is stored in logic. Following the spark and the recovery time, the stored value less an adjustable amount defines the new excitation level. This adjustment called setback, has a range labeled minimum to maximum. The range is adjustable from 1/2 percent to 10 percent of rated current. Normally a mid range setting is desirable.

Spark Intensity - This provides a vernier adjustment for the spark detection circuit's sensitivity to a spark. Thus occasional light sparks which may be self-quenching will not cause the control to remove excitation. A spark is detected by monitoring a sudden drop in the long term value of secondary voltage. On spark detection, the spark light will flash. For most precipitator processes, this adjustment is normally set in the full CCW position to provide maximum sensitivity to sparks.

7.0 PREVENTIVE MAINTENANCE GUIDE

Maximum trouble-free service from Precipitator controllers requires periodic inspection, preventive maintenance, and periodic cleaning. A definite schedule should be maintained for inspection, the frequency depending upon operating conditions.

Preventive maintenance activity should then be established as the result of periodic inspection.

In these routine inspections, four basic categories of deteriorating influences should be kept in mind:

a. The effect of foreign material and chemicals in the atmosphere: fly ash, cement dust, wood fibers, corrosive chemicals, and residues.

b. Mechanical wear and fatigue on all moving parts.

c. Heat

d. Loose joints and connections.

The following are some specific recommendations :

1. During installation and start up, environmental conditions are generally the worst. A thorough cleaning of the controller is recommended.

2. Check for cleanliness generally, but particularly for accumulation of foreign materials obstructing the inlet and outlet ventilation. Poor ventilation will cause decreased component life and overtemperature causing control trips.

3. Check for build up of dust or dirt which would reduce any forced air circulation.

4. Fans should be inspected periodically to verify that shaft rotates freely when power is off.

8.0 CONTROLLER ELEMENTARY

ELEMENTARY INDEX

Fig. 8.1 – Local and remote metering (kV, primary volts, spark rate)

– Control adjustment (spark intensity, spark rate, setback)

Fig. 8.2 – Local and remote selector switch (Auto/manual, local/remote)

– Control adjustments (manual, remote manual, current limit)

Fig. 8.3 – Alarm lights and trips (undervoltage, overvoltage, T/R overtemperature)

– Alarm contacts for remote use (overvoltage, undervoltage, and T/R overtemperature)

– Undervoltage and Overvoltage adjustment

Fig. 8.4 – Alarm lights and trips (S. C. R. overtemperature, overcurrent)

– Alarm contacts for remote use (S. C. R. overtemperature and overcurrent)

– Local and remote metering (primary current)

Fig. 8.5 – Load current feedback from T/R

– Local and remote metering (load current)

– Control adjustment (recovery)

Fig. 8.6 – Local and remote ON-OFF/RESET push buttons and lights

– ON-OFF/RESET contacts for remote use

Fig. 8. 7 – Power circuit (CB, fuses, MI contactor, SCR's, reactor)

– Fans, control power transformers

– SCR regulator and CB auxiliary contacts for remote use.

Fig. 8.8 – ±15 Volts dc and 90 Volts dc power supply

– ±15 Volts dc, 90 Volts dc and common panel wiring

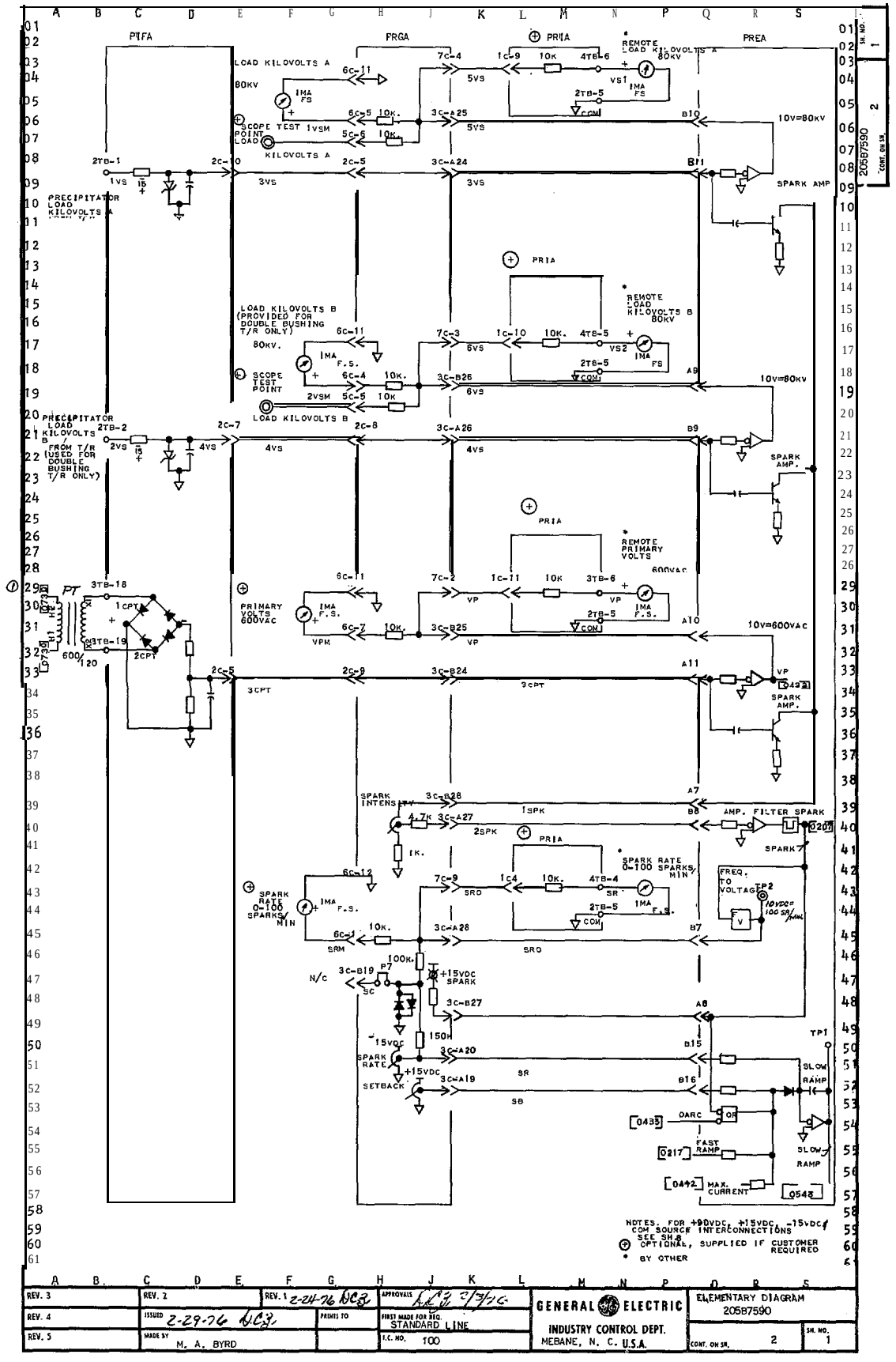


Fig. 8.1 *Local* and remote mefering

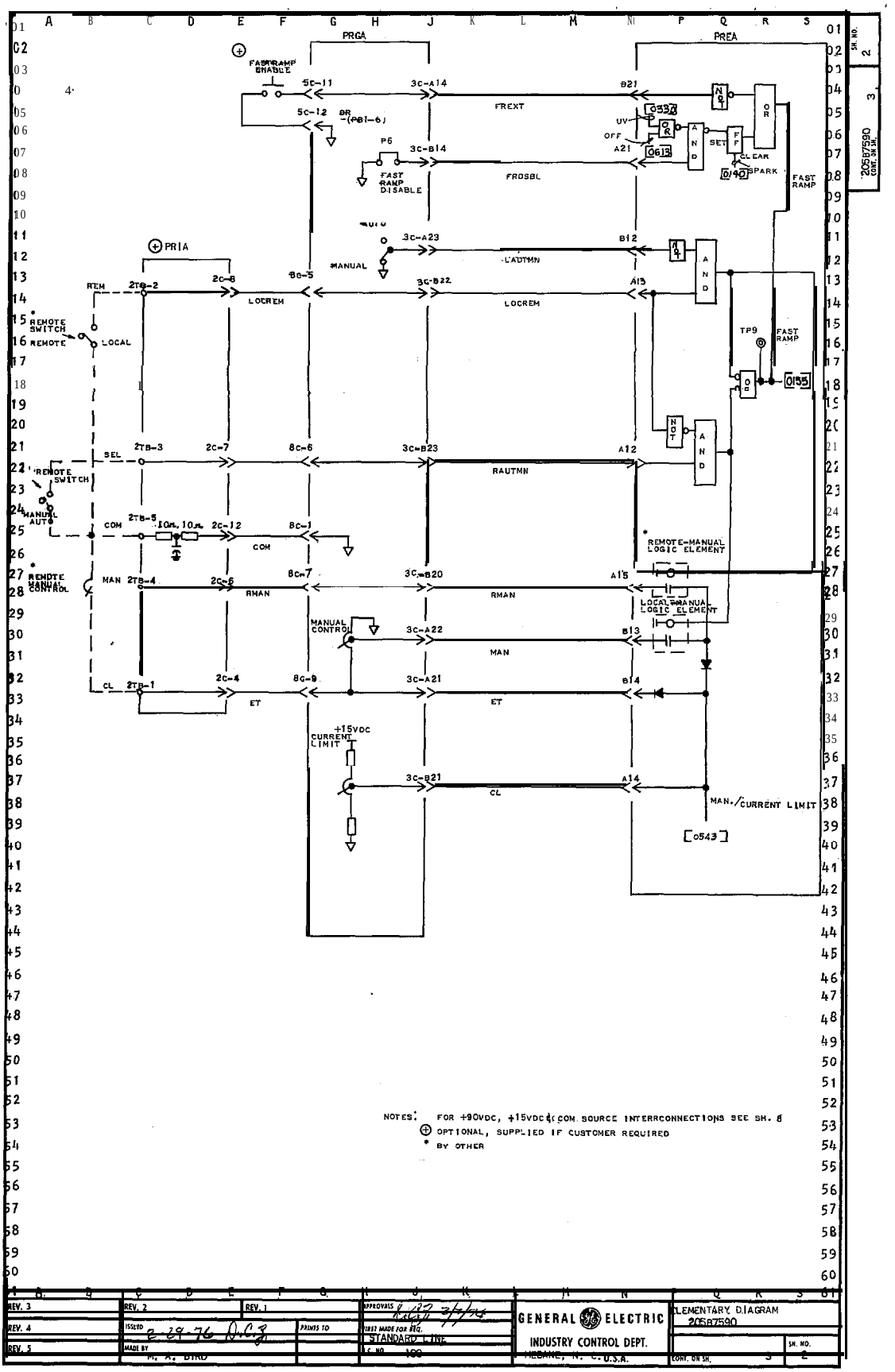


Fig. 8.2 Local and remote selector switch

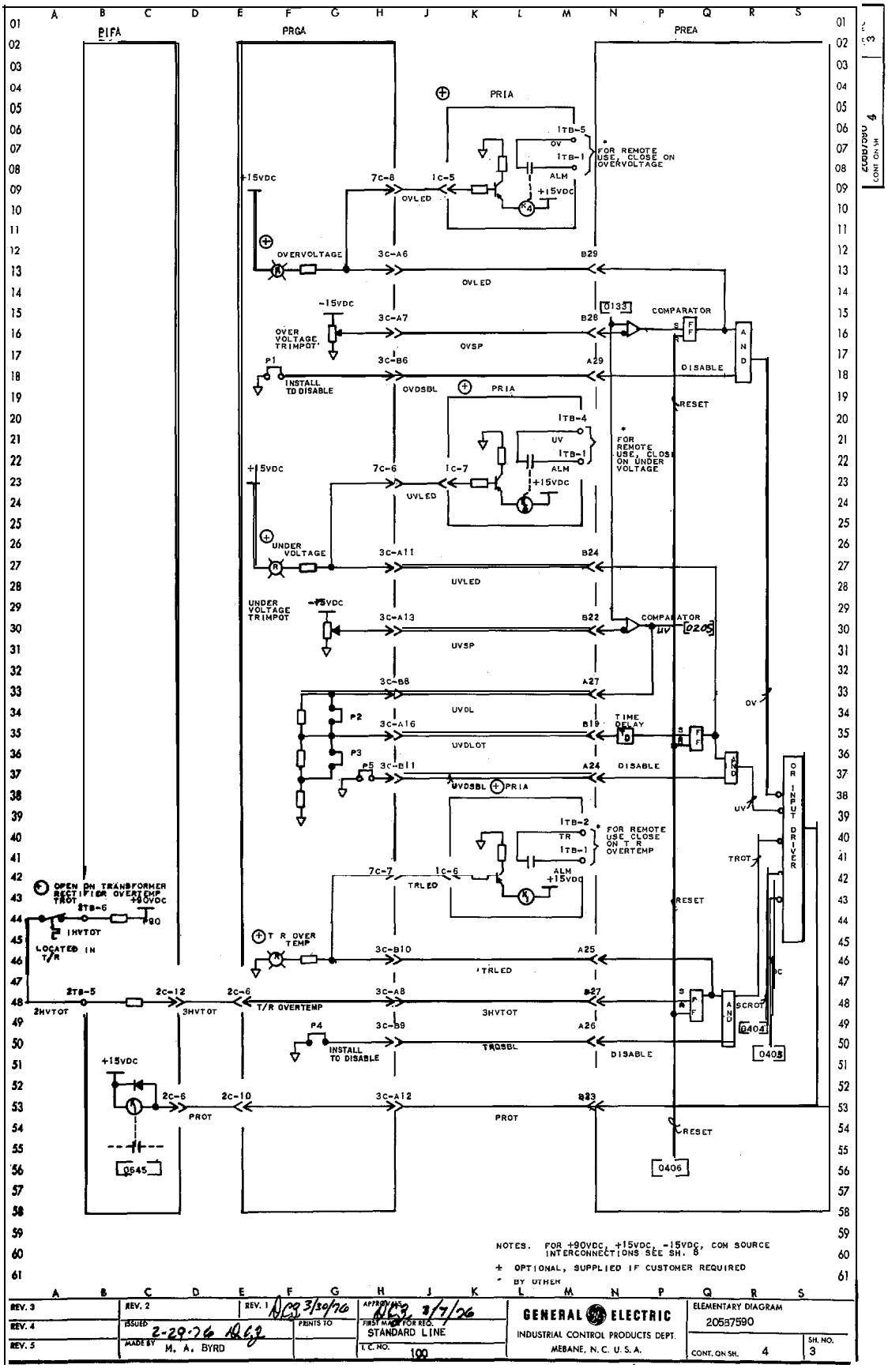


Fig. 8.3 Alarm *lights* and trips

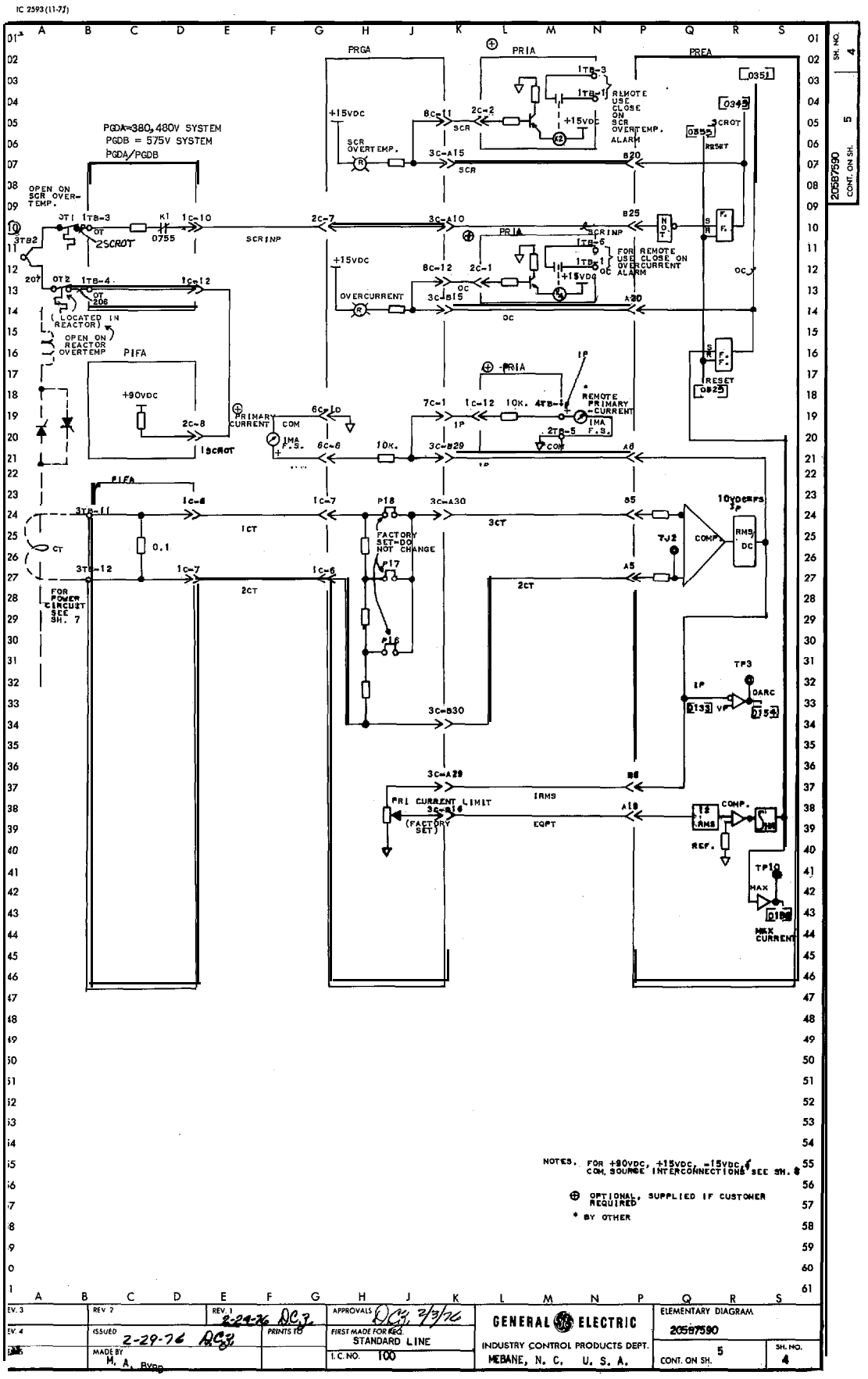


Fig. 8.4 Alarm *lights* and trips

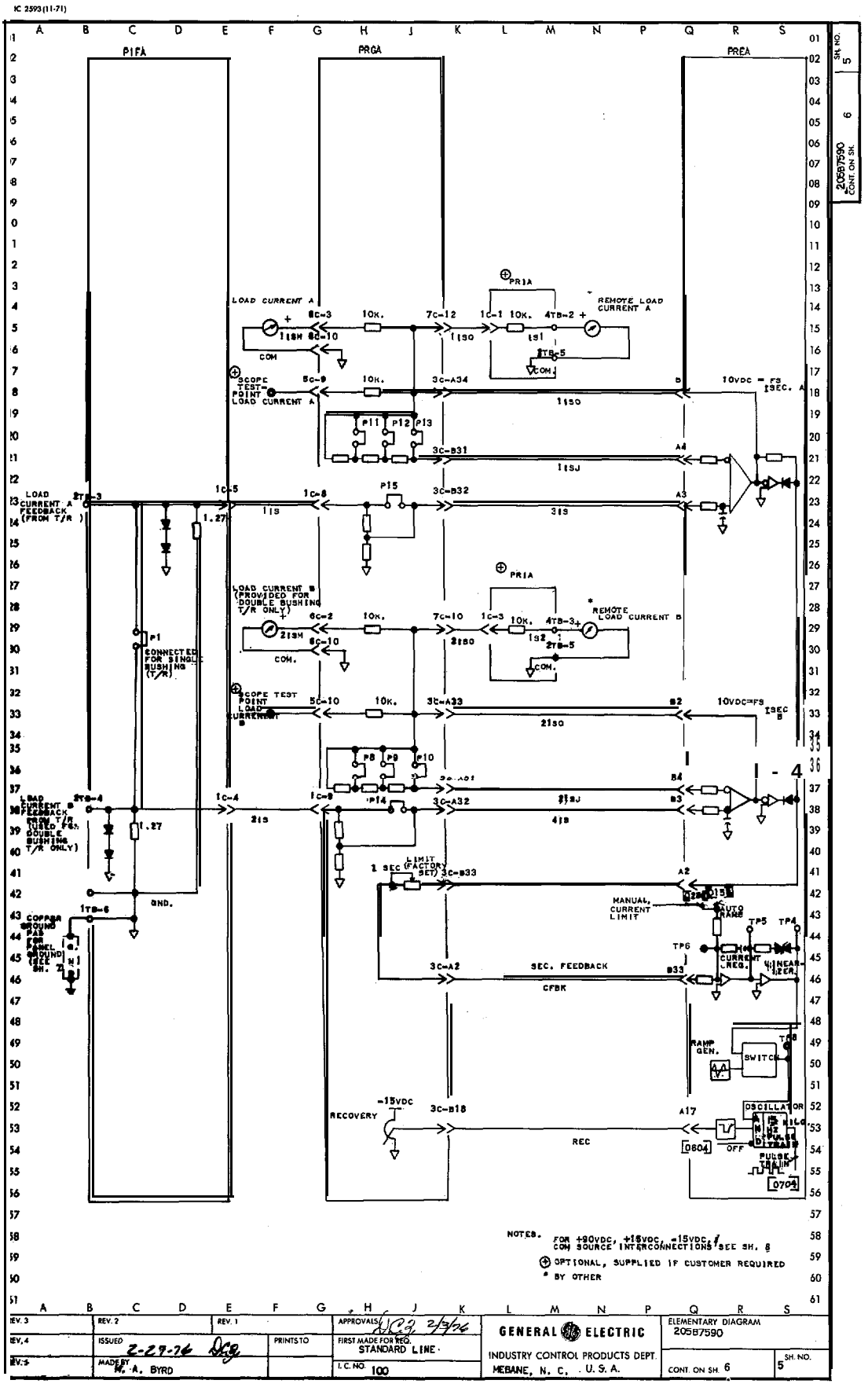


Fig. 8.5 *Load current feed back from T/R*

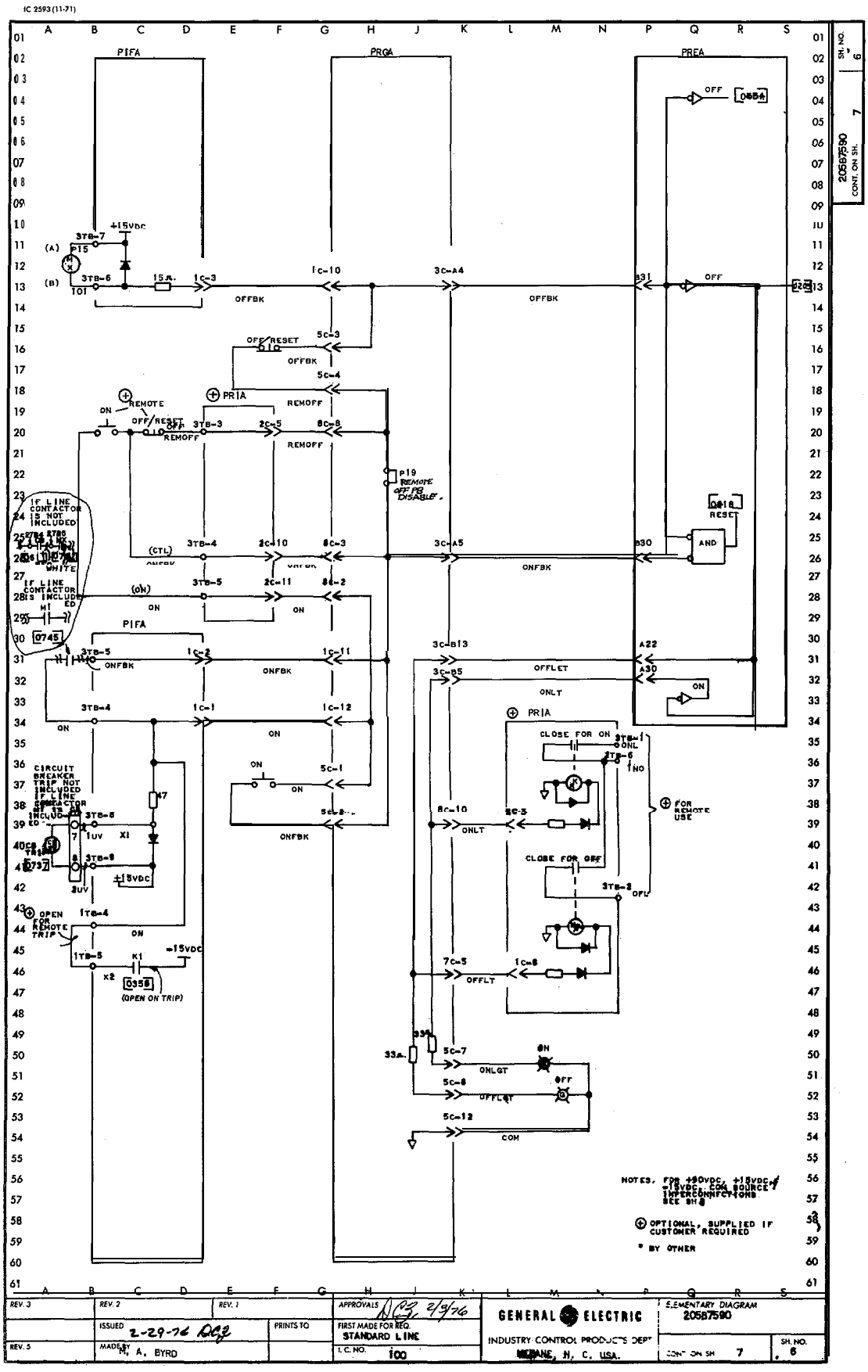


Fig. 8.6 local and remote ON-OFF/reset push buttons and fights

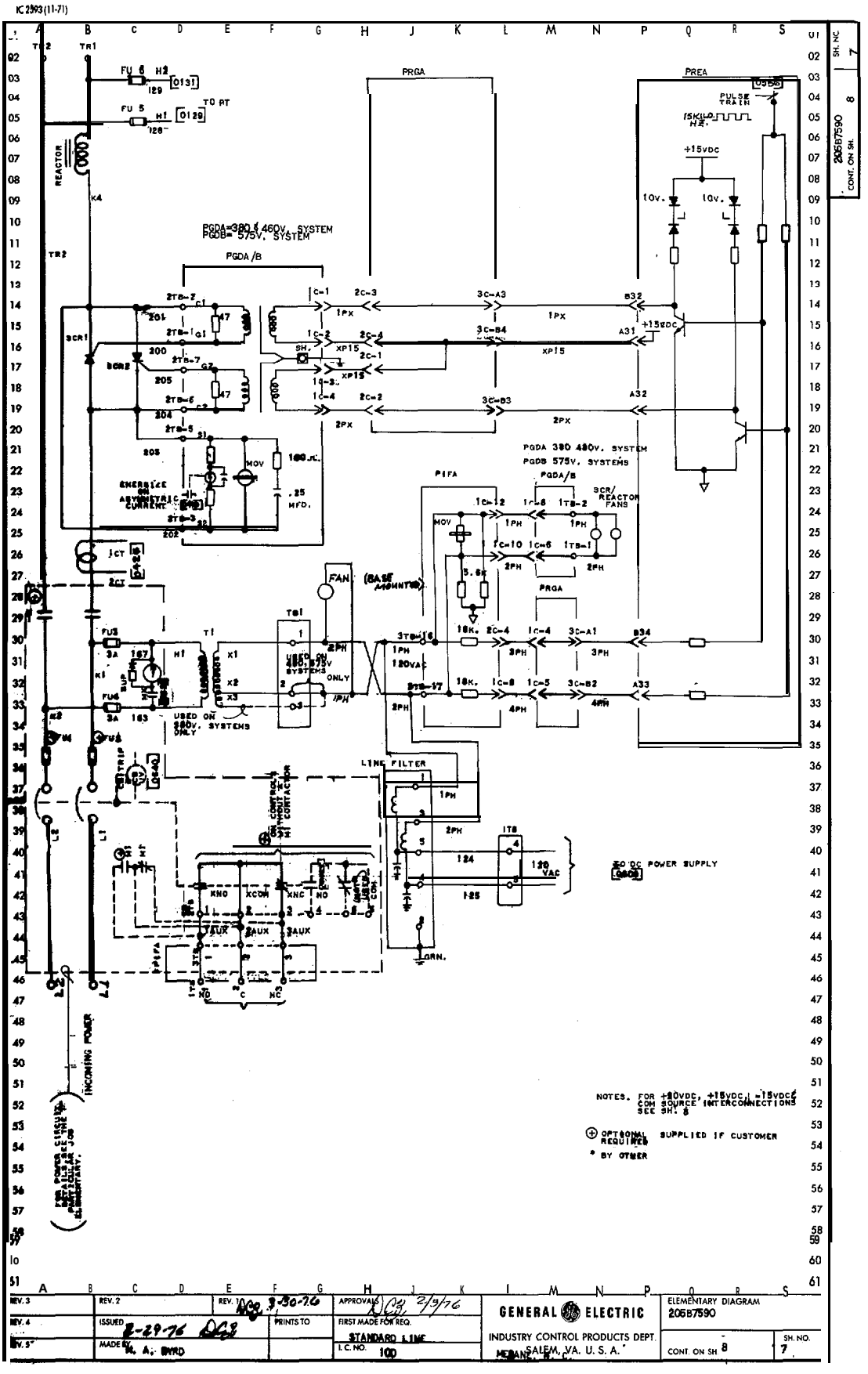


Fig. 8.7 Power circuit

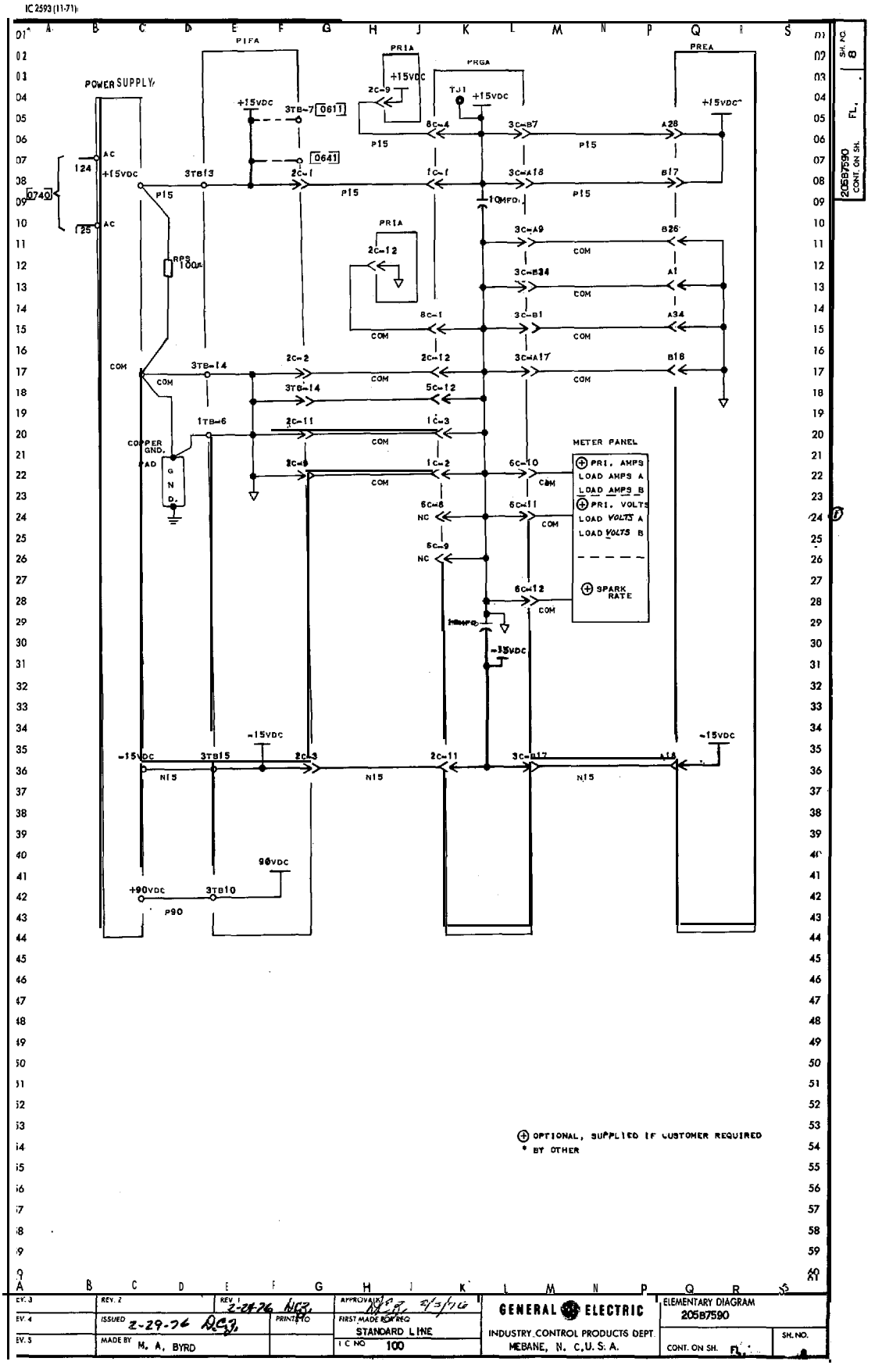


Fig. 8.8 Power supply and common panel wiring

9.0 TROUBLESHOOTING PROCEDURE

NOTE:

- **Components** *are* **not always** *interchangeable from* **panel to panel.**
- **Panels with** *different* **nameplate** *current ratings* **may** *require* **different SCR** *ratings*, **CT ratios,** *reactor* **ratings, CB** *frame* **and trips,** *contactor* **sizes, meter** scales and **PRGA** *programming* **and** *adjustments*.
- **Panels of the** *same* **nameplate current** *ratings*, **voltage** *ratings*, *and options generally* **use interchangeable** *components*.
- *Connector* **and** *terminal board* **locations** *are* **shown in Figures 2.1,** *2.2*, **2.3,** *2. 6, 2. 7 and 5.2.*

TROUBLE	CAUSE	REMEDY
9.1 POWER MAGNETICS AND ON/OFF CIRCUITS		
Circuit Breaker immediately trips on closure.	Short circuit line to line or line to ground.	1. With the CB open, check load circuit using an ohmmeter for short circuits.
	Circuit breaker under voltage trip coil "CBUV" is not energized. (Provided on controllers without an Ml contactor) Fig. 8. 6	1. Provide 24 Volts dc to "CBUV" terminals by: a. Verifying remote trip points 1TB5 and 1 TB4 on the PIFA card are connected. b. Verifying +30 Volts dc between PIFA 3TB9 and PIFA 3TB8. (If no Ml contactor is included). 2. Depress OFF/RESET PB to reset all alarms to energize K1 relay on the PIFA card. (Reference 6.0, Alarm Circuit) 3. Verify "CBUV" coil resistance between PIFA 3TB9 and PIFA3 TB8. (If no Ml contactor is included).
	SCR cell shorted	1. With the CB off, use a VOM on the R scale to measure across the SCR's. Measure resistance between the top SCR power terminal and the SCR power terminal connecting to the reactor. Resistance of 100 ohms or less indicates an SCR is shorted and should be replaced. The SCR heat sink assembly is 'bolt in" and front removable.
Contactor *(Ml) will not 'pick up"	Line voltage is not being applied to contactor coil when "ON" PB is depressed. Reference Fig. 8.6.	1. Verify remote trip points ITB5 and ITB4 on the PIFA card are connected together. 2. Depress OFF/RESET to reset all alarms to energize K1 on the PIFA card. (Reference 6. 0 Alarm Circuit.)

*Optional

TROUBLE	CAUSE	REMEDY
Contactor *(Ml) will not "pick up" (cont'd)		3. Measure line voltage (460 Volts ac) across MX contact from MX-167 to MX-163 with MX de-energized. 4. Verify relay MX "picks up" when "ON" is depressed.
	Contactor coil is open	1. With power off, measure coil resistance for an open coil. (Faulty: Greater than 100 ohms). 2. Replace coil if open.
"OFF" light is not lit when control is "OFF."	Lamp is incorrect or burned out. Circuit breaker must be "ON" for controls with an Ml contactor.	1. 12 Volts dc filament requires exact replacement. Verify bulb filament voltage and measure 12 Volts dc across filament terminals.
Fans are not "ON" when circuit breaker is closed. (If an Ml contactor is included, the CB must be closed. If not included, fans should run before CB is closed.)	120 volts ac power from secondary of transformer (T1) is not available. Reference Fig. 8.6.	1. Verify line voltage is present between FU3 and FU4 bottom terminals. 2. Replace FU3 and FU4 if open.
"ON" light is lit while "ON" PB is depressed but goes out when released. (Reference Fig. 8-6).	MX relay does not pick up.	1. Verify the remote OFF/RESET PB is connected between points 3TB3 and 3TB4 on the *PRIA card.
	MX relay 'picks up" when "ON" PB is depressed. But drops out when "ON" PB is released.	1. Verify the MX seal circuit around the "ON" PB by verifying the CB interlock is closed and the MX interlock closes when on is depressed. The seal circuit is between PIFA 3TB4 and PIFA 3TB5. If an M1 contactor is included, the seal circuit consists of an Ml interlock only, which closes when Ml picks up.
	MX relay "picks up" and stays up until OFF/RESET is depressed. (If an M1 contactor is included, Ml picks up and stays up,)	1. Verify wiring for ONLT and OFFLT signals. Reference Fig. 8.6. 2. Replace the PREA CARD.

9.2 ALARM CIRCUIT

Reference Fig. 8.3, 8.4, and 8.6 T/R Overtemperature alarm will not reset.	Remote TROT interlock is not closed.	1. An overtemperature condition exist in the T/R: Verify remote normally closed interlock (TROT) is connected between PIFA 2TB5 and PIFA 2TB6 by measuring continuity between these points.

*Optional

TROUBLE	CAUSE	REMEDY
T/R Overtemperature alarm will not reset. (cont'd)	Reset logic from OFF/RESET PB is inoperative. (Reference Fig. 8.6, Line 26).	1. With a VOM measure 15 ±3 Volts dc from PRGA IC11 to ground. When the OFF/RESET PB is depressed the voltage should go to 0 Volts dc for a common reset logic input for all alarms. 2. Replace the ribbon cable connecting the PREA to the PRGA card. 3. Replace the PREA card.
Overcurrent alarm will not reset.	Reset logic from OFF/ RESET PB inoperative. (Reference Fig. 8.6. Line 26).	1. With a VOM measure 15 ±3 Volts dc from PRGA IC11 to ground. When OFF/RESET PB is depressed, the voltage should go to 0 volts dc for a common reset logic input for all alarms. 2. Replace the ribbon cable between the PREA to the PRGA card. 3. Replace the PREA card.
SCR overtemperature alarm will not reset.	Improper ventilation	1. Verify both SCR fans run. 2. Verify no obstruction exists which might prevent proper air flow over the SCR heat sink fins.
	Overtemperature alarm signal to PREA card.	1. Verify 90 Volts dc exist between PGDA 1 TB4 and ground. 2. Verify SCR and reactor thermostats are closed by checking continuity between 1TB4 and 1C10 on the PGDA card. 3. Replace the PREA card.
*Overvoltage alarm will not reset	An overvoltage exists at the primary of T2. (Reference Fig. 8.6 line 26.)	1. Excessive line voltage at transformer T2. 2. Improper overvoltage setting. (See overvoltage listed under Paragraph 6.1).
	Reset logic from the OFF/ RESET PB inoperative. (Reference Fig. 8.6. line 26).	1. With a VOM measure 15 ±3 Volts dc from PRGA IC11 to ground. When the OFF/RESET PB is depressed the voltage should go to 0 Volts dc for a common reset logic input for all alarms. 2. Replace the ribbon cable connecting the PREA to the PRGA card. 3. Replace the PREA card.
*Undervoltage alarm will not reset	An undervoltage exists at the primary of T2. (Reference Fig, 8.1.)	1. A ground exists in the precipitator. 2. Low voltage exists at the primary of transformer T2.

*Optional

TROUBLE	CAUSE	REMEDY
*Undervoltage alarm will not reset (cont'd)		3. Improper undervoltage setting. (See undervoltage listed under Paragraph 6.1).
	Reset logic from the OFF/ RESET PB inoperative.	1. With a VOM measure 15 ±3 Volts dc from PRGA IC11 to ground. When the OFF/RESET PB is depressed the voltage should go to 0 Volts dc for a common reset logic input for all alarms. 2. Replace the ribbon cable connecting the PREA to PRGA card. 3. Replace the PREA card.

9.3 CONTROL ADJUSTMENTS

Reference Fig. 8.1, 8.2, and 8.3. Manual adjustment does not vary primary voltage or current.	Auto-Manual switch must be *in* manual position.	1. Select manual position.
	Remote switch for *LOCAL-remote selection must be in local position.	1. Select local position. 2. Verify 2TB2 on *PRIA card is not connected to common by measuring continuity to ground with PREA ribbon cable disconnected. (2TB2 to corn. would disable local manual adjustments).
	Component failure.	1. Replace PREA ribbon cable. 2. Replace PREA card. 3. Replace PGDA/B card. 4. Replace PRGA card.
	No SCR gate pulse train signal to turn on SCR's.	1. With CB and *contactor closed, monitor with a VOM 0 to 15 Volts dc across the primary of the SCR pulse transformers. Monitor points for SCR1 are PRGA 2C 4 to PRGA 2C 3. For SCR 2 monitor PRGA 2C 1 to PRGA 2C 2. With a manual adjustment C. C. W. the voltage should be less than 1 Volts dc for both SCR's. With the adjustment increased to mid position, the voltage should increase to 4-8 Volts dc and be approximately equal for each SCR. If voltages are correct, first replace PGDA/B card. Next replace SCR Heat sink assembly.

*Optional

TROUBLE	CAUSE	REMEDY
Current limit adjustment is inoperative	Improper load current feedback	1. Verify T/R secondary current feedback is connected to: PIFA 2TB3, for single bushing T/R feedback; double bushing feedback connects to PIFA 2TB3 and PIFA 2TB4.
	Failed component.	1. Replace PREA ribbon cable. 2. Replace PREA card. 3. Replace PRGA card.
Recovery adjustment inoperative	Failed component.	1. Replace PREA ribbon cable. 2. Replace PREA card. 3. Replace PRGA card.
Setback adjustment inoperative	Failed component.	1. Replace PREA ribbon cable. 2. Replace PREA card. 3. Replace PRGA card.
Spark intensity inoperative	Control not sensing a spark.	1. Sparking must occur in the Precipitator and the "spark" light or the control must flash.
	Failed component.	1. Refer to the precipitator manufacturer to determine if light sparking is anticipated. 2. Replace PREA ribbon cable. 3. Replace PREA card. 4. Replace PRGA card.
Manual/Auto Switch inoperative	*Remote switch for local-remote selection in remote position	1. Select local position. 2. Verify 2TB2 on PRIA card is not connected to common by measuring continuity to ground with the PREA card ribbon cables disconnected.
	Component failure	1. Replace PREA ribbon cable. 2. Replace PREA card. 3. Replace PRGA card.
Spark rate adjustment inoperative	Auto/Manual switch must be in auto position	1. Select auto position. 2. Control may be operating at current limit which may be below the sparking level.
	Remote switch for local-remote selection is in the remote position.	1. Select local position. 2. Verify 2TB2 on the *PRIA card is not connected to common by measuring continuity to ground with the PREA ribbon cables disconnected. 3. Replace PREA ribbon cable. 4. Replace PREA card. 5. Replace PRGA card,

*Optional

9.4 METERING

Reference Fig. 8.1.
8.4, and 8.5.

NOTE: ***Standard meters are 1.0 milliamp movement electronically driven by a O-10 volts dc signal thru a prevision 1 OK ohm resistor to provide 1.0 milliamp full scale output. Meter signals are short circuit protected, thus*** may ***be disconnected or shorted to ground.***

Remote meters are 1.0 milliamp movement and uses the same O-10 volts dc signal thru a precision 10K ohm resistor on the *PRIA card. The following troubleshooting guide can also be used for remote meter problems.

TROUBLE	CAUSE	REMEDY
*Primary voltmeter inoperative Reference Fig. 8.1.	Defective meter	1. Replace meter.
	No input or improper input signal to meter.	1. Verify FU5 and FU6 are good. 2. With controller "ON" and primary current flowing: measure O-480 volts ac to the primary of T2 with a VOM between FU5 and FU6. 3. At the secondary of PT measure 1/5 this voltage; (T2 = 5:1 ratio) between PIFA 3TB18 and PIFA 3TB19. 4. At the PRGA 2C9 to common measure approximately 1/12 the voltage measured in step 3 above. (6-7 volts dc) 5. Replace the PREA ribbon cable. 6. Replace the PREA card.
*Primary current meter inoperative Reference Fig. 8.4	Defective meter	1. Replace meter.
	No input or improper input signal to meter.	1. Measure ac voltage with a VOM across the terminals of the CT. Voltage = $\frac{\text{(0.1 ohm) x (Primary amps)}}{\text{CT Ratio}}$ 2. Measure 0.375 ac voltage between PRGA IC6 to PRGA IC7. Fullscale current equals 0.375 vac. 3. Replace the PREA ribbon cable. 4. Replace the PREA card.
Load current meter(s) Reference Fig. 8.5	Defective meter(s)	1. Replace meter(s).
	No input or improper input signal to meter(s)	1. Verify load current feedback from T/R is connected to PIFA 2TB3 for single bushing and PIFA 2TB3 and 2TB4 for double bushing.

*Optional

TROUBLE	CAUSE	REMEDY
Load current meter(s) Reference Fig. 8. 5 (cont'd)		2. Verify P1 shorting screw located in the upper right of PIFA card is installed for a single bushing T/R and not installed for a double bushing T/R. 3. Replace the PREA ribbon cable. 4. Replace the PREA card. 5. Replace the PIFA card.
Load kilovolt meter(s) Reference Fig. 8.1	Defective meter(s)	1. Replace meter(s).
	No input or improper input signal to meter(s)	1. Verify secondary kilovolts feedback (1.0 milliamp full scale signal) from the T/R is connected to PIFA 2TB1 for a single bushing T/R and PIFA 2TB1 and PIFA 2TB for a double bushing T/R. 2. With the CB off, continuity check wiring between PIFA 2C 10 to PRGA 2C5. Also check PIFA 2C 7 to PRGA 2C8 for a double bushing T/R. 3. Replace the PREA ribbon cable. 4. Replace the PREA card.
Spark rate meter inoperative Reference Fig. 8.1	Defective meter	1. Replace meter.
	No input or improper input signal to meter.	1. Replace the PREA ribbon cable. 2. Replace PREA card. 3. Replace PRGA card.

9.5 VOLTAGE MEASUREMENTS

NOTE: ***Voltage measurements specified are plus or minus 10 percent unless otherwise indicated. For dc measurements, the first point in the tabulation is positive. Measurements may be made with a VOM.***

9.5.1 Incoming line

Circuit	Monitor Point	Voltage
Circuit Breaker (CB)	CB-L1, CB-L2	Reference Controller Nameplate (Fig. 2.6) (380, 460, 575 Volts ac) ± 5 percent
Control Transformer (TI)	FU3, FU4	

9.5.2 Control Voltage

Circuit	Monitor Point	Voltage
Device Panel Fan (FAN)	TB1-1, TB1-2	120 Volts ac
Power Supply (PS)	PS-124, PS-125	120 Volts ac
SCR Phase Reference (PRGA)	PRGA 1C4, PRGA 1C5	120 Volts ac
Line Filter (FLT) Input	TB1-1PH, TB1-2PH	120 Volts ac
Line Filter (FLT) Output	TB1-124, TB1-125	120 Volts ac
SCR/Reactor Fans (PGDA/B Card)	PDGA/B 1TB2 PGDA/B 1TB1	120 Volts ac

NOTE: ***PGDA for 380 & 460 volt systems, PGDB for 575 volt systems.***

9.5.3 Electronic Power

Circuit	Monitor Point	Voltage
Power Supply (PS)	PS-P15, PS-COM	+ 15 Volts dc
Power Supply (PS)	PS-P15, Case ground	+ 15 Volts dc
Power Supply (PS)	PS-COM, PS-N15	- 15 Volts dc
Card (PIFA)	PIFA-3TB13, PIFA-3TB14	+ 15 Volts dc
*Card (PRIA)	PRIA 2C9, PRIA 2C12	+ 15 Volts dc

9.5.4 DC Control Power

Circuit (Device Designation)	Monitor Point	Voltage
Power Supply (PS)	PS-P90, PS-COM	+ 90 Volts dc
Card (PIFA)	PIFA 3TB10, PIFA 3TB14	+ 90 Volts dc
*Remote T/R overtemperature contact		
closed (normal condition)	PIFA 2TB6, PIFA 2TB5	0 Volts dc
open (alarm condition)	PIFA 2TB6, PIFA 2TB5	+ 90 Volts dc
Control Relay (MX)	MX-P15, Case Ground	+ 15 Volts dc
MX-Energized (MX)	MX-P15, MX-101	+ 24 ± 4 Volts dc
MX-DE -Energized (MX)	MX-P15, MX-101	0 Volts dc
ON Light-Control "ON" (PBI)	PBI-ONLTG, PBL-COM	+ 12 Volts dc
OFF Light-Control "OFF" (PB2)	PB2-OFFLTG, PB2-COM	+ 12 Volts dc
ON PB (PB 1)	PB1-COM, PB1-ON	- 15 Volts dc

Circuit (Device Designation)	Monitor Point	Voltage
OFF/Reset PB-Control "OFF" (PB2)	PB2-OFFLTG, P2-COM	+ 15 Volts dc
OFF/Reset PB-Control "ON" (PB2)	PB2-COM, PB2-ONFBK	- 15 Volts dc

9.5.5 The Following Voltage Vary According to Conducting Angle

Circuit (Device Designation)	Monitor Point	Voltage in VAC
Transformer Primary (PT)	FU5-2CPT, FUG-ICPT	0 to line voltage
Transformer Secondary (PT)	PIFA 3TB18, PIFA 3TB19	0 to $\frac{\text{line voltage}}{5}$
Current Transformer (CT)	CT-1CT, CT-2CT	0 to $\frac{\text{line amps x 0.1 ohms}}{\text{CT Ratio}}$

* Optional

9.5.6 Metering (Standard Controller)

NOTE: All metering uses a 10 volts dc signal thru a 1 OK OHM resistor for full scale deflection. All meter movements are 1.0 milliampere dc. *If a* meter *lead is disconnected for panel metering or *remote metering, a 0 to 10 volts dc output signal thru a 10K resistance can be measured to common.*

9.5.7 Scope Monitor Points

(Reference Elementary Diagram Fig, 8.1, and 8.5)

Circuit (Device Designation)	Monitor Point	Voltage
Load Kilovolts A (Card)	PRGA 5C6, Case Ground	0 to 10 Volts dc
Load Kilovolts B (Card)	PRGA 5C5, Case Ground	0 to 10 Volts dc
Load Current A (Card)	PRGA 5C9, Case Ground	0 to 10 Volts dc
Load Current B (Card)	PRGA 5C10, Case Ground	0 to 10 Volts dc

10.0 RENEWAL PARTS

The renewal parts list is made for each job since controller requirements and parts list varies from job to job. A complete renewal parts list can be obtained through your General Electric Sales Office.

11.0 ORDERING INFORMATION

A unique catalog number will be given every controller depending on rating and option content. In order to promptly process your order, the following information about your controller is essential.

11.1 PRIMARY VOLTAGE RATING

☐ 460 Volts, 60 HZ (Standard)

☐ 380 Volts, 50 HZ

❑ 575 Volts, 60 HZ

11.2 SECONDARY VOLTAGE RATING

_____ Kilovolts dc average (45 kV to 65 kV, 5 kV increments)

11.3 SECONDARY CURRENT RATING

_____ Milliampere dc average, 100 mA to 2500 mA, (100 mA increments)

11.4 AMBIENT TEMPERATURE

☐ 40 C (Standard)

❑ 50 c

11.5 TRANSFORMER ARRANGEMENT

☐ Single Bushing, Full Wave (Standard)

☐ Double Bushing, Double Half Wave

11.6 HORIZONTAL BUS

☐ No Bus (Standard)

☐ S-phase Horizontal Bus

(Attach sketch showing multicase lineup, bus size and phase connection)

11.7 SHORT CIRCUIT CAPACITY

☐ 14,000 Amperes, Symmetrical (Standard)

_____ Amperes. (14, 18, 22, 25, 30, 42, 65,100) kiloamps

11.8 PRIMARY METERING (options)

☐ Primary voltmeter

c l Primary ammeter

☐ Spark Rate Meter

11.9 FAULT/TRIP OPTIONS

Transformer Overtemperature

❑ Trip and Indicate

❑ Indicate Only

Undervoltage/Overvoltage

- ☐ Trip and Indicate
- ☐ Indicate Only
- ☐ 1/2 Minute Setting
- ☐ 1 Minute Setting (Standard)
- ☐ 2 Minute Setting

11.10 ENCLOSURE

- ☐ NEMA 1A (Standard)
- ☐ NEMA 12

11.11 POWER CIRCUIT

- ☐ Circuit Breaker Only (Standard)
- ☐ Circuit Breaker and Contactor
- ☐ Auxiliary Contact (1NO & INC)

11.12 REMOTE OPTION

- ☐ No (Standard)
- ☐ Yes

Specify Requirements

Remote Alarms
- ☐ Remote T/R Overtemperature Alarm
- ☐ Remote SCR Over-temperature Alarm
- ☐ Remote UV Alarm
- ☐ Remote OV Alarm
- E l Remote Over-current Alarm

Remote Controls
- ☐ Local/Remote Control Switch
- ☐ Remote Auto/Manual Switch
- ☐ Remote Manual Control
- ☐ Remote ON/OFF Lights
- ☐ Remote ON/OFF PB's

Remote Metering
- ☐ Remote Metering PRI Volts
- ☐ Remote Metering PRI Amperes
- ☐ Remote Metering SEC. Kilovolts
- ☐ Remote Metering SEC Amperes
- ☐ Remote Metering Spark Rate

11.13 TRANSFORMER NAMEPLATE INFORMATION

General Electric Requisition Number

Attach all other applicable information.

GENERAL ELECTRIC COMPANY, U.S.A.
INDUSTRIAL CONTROL DEPARTMENT

GENERAL@ ELECTRIC

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